

SECTION A

ANSWER ALL QUESTIONS

1. The rigidity and shape of the bacterial cell is due to the:
A. Cell membrane
B. Peptidoglycan layer
C. The lipopolysaccharide
D. The teichoic acid layer
2. Which of the following is found in both Gram positive and Gram negative cell walls?
A. An outer membrane
B. Teichoic acid
C. Peptidoglycan
D. Lipopolysaccharide
3. Media used to cultivate bacteria according to their differential colored colonies is called
A. Indicator media
B. Transport media
C. Enriched media
D. Anaerobic media
4. MacConkey's agar and TCBS agar are examples of
A. Indicator media
B. Transport media
C. Enriched media
D. Anaerobic media
5. Which of the following is not applicable to culture media?
A. Is used to study environmental and nutritional factors
B. Is a convenient way of separating bacteria in mixtures
C. Is a reliable way of estimating bacterial viable counts
D. Culturing bacteria is an initial step in studying its morphology
6. Which of the following is true for bacterial conjugation?
A. The donor cell do not provide a mobilizable genetic element during conjugation
B. Conjugation is a mechanism of horizontal gene transfer
C. It sometimes involve cell to cell contact
D. Both the pilus and the genetic material migrates to the recipient during conjugation
7. Which one of the following culture media is used on fastidious organisms that require defined known quantities of specific nutrients?
A. Nutrient broth
B. Nutrient Agar
C. Complex Media
D. Defined Media
8. Endospores:
A. Represent a dormant cell stage resistant to heat, dessication, and many chemicals.
B. Allow cells to float or position themselves in a water table.
C. Are formed in conidia that develop in fruiting bodies arising out of a mycelial mass.
D. Are made up predominantly of water.
E. Function in cellular replication.

*9. The solidifying agent used most successfully in bacterial nutrition media

- A gelatin
- B peptone

- C agar agar
- D starch

10. Prokaryotic cells divide by a process known as

- A. Conjugation
- B. Mitosis

- C. Binary fusion
- ☒ D. Binary fission

11. In the growth curve of a bacterial population, the bacteria are rapidly increasing in number in the

- A. Lag phase
- ☒ B. exponential/ log phase

- C. stationary phase
- D. decline phase

12. Most Gram-negative bacteria have a lipopolysaccharide (LPS) which is found in

- A. plasma membrane
- B. inner membrane
- ☒ C. outer membrane
- D. periplasmic space

13. The sterilization temperature, pressure and time for autoclave is

- ☒ A. 121°C, 15psi, 15 minutes
- B. 160°C, 20psi, 30 minutes

- C. 111°C, 15psi, 25 minutes
- D. 120°C, 15psi, 15 minutes

14. The following stain(s) is used to detect *Mycobacterium* species:

- ☒ A. Acid Fast
- B. Capsule Stain

- C. Gram Stain
- D. Simple Stain

15. One of the following is a β -lactam antibiotic

- A. Suphanamides
- ☒ B. Novobiocin

- C. Aminoglycosides
- D. Cephalosporins

16. Which dye is used as a mordant during the gram staining procedure?

- A. Alcohol
- ☒ B. Iodine
- C. safranine
- D. carbol

*17. One of the following samples should be analyzed apart from CSF when a patient has a central nervous system infection

- A. Blood
- B. Stool

- ☒ C. Sputum
- D. Urine

18. Which of the following diseases was used as the basis for Koch's postulates?

- ☒ A. Syphilis

- ☒ B. Small pox

- C. Bacterial meningitis

- D. Anthrax

19. Nutrient material prepared for microbial growth is called

- A. Culture

- ☒ B. Culture Medium

- C. Inoculum

- D. None of the above

is an important component of the culture media because

- A. Not degraded by microbes
- B. Solidifies at $\sim 40^{\circ}\text{C}$ & remains solid at room temperature
- C. All of the above
- D. None of the above

DNA replication occur in _____ direction

- A. $5' \rightarrow 3'$
- B. $3' \rightarrow 5'$
- C. All of the above
- D. None of the above

1. Which of the following enzymes involved in DNA replication is responsible for the formation of okazaki fragment?

- A. DNA polymerase III
- B. DNA polymerase I
- C. RNA polymerase
- D. Topoisomerase

2. The template for the strand synthesized in fragment is _____

- A. Leading
- B. Lagging
- C. Complementary
- D. Template

24. The default position of a repressible gene is _____

- A. Off
- B. On
- C. On/Off
- D. Operon

25. Operon consist of the following except _____

- A. Operator
- B. Promoter
- C. Structural genes
- D. RNA Polymerase

26. In inducible gene the repressor protein is always _____

- A. Active
- B. Inactive
- C. All of the above
- D. None of the above

27. Which of the following is not a method of horizontal gene transfer?

- A. Binary fission
- B. Conjugation
- C. Transduction
- D. Transformation

28. The site at which peptide bond is formed is _____

- A. E-site
- B. P-site
- C. A-site
- D. T-site

29. Which of the following enzymes initiate transcription process?

- A. DNA polymerase III
- B. DNA polymerase I
- C. RNA polymerase
- D. Topoisomerase

30. Base triplet portion of the tRNA is termed _____

- A. Codon
- B. Anticodon
- C. Nonsense Codon
- D. Sense Codon

31. A satellite is a subviral agent composed of nucleic acid that depends on the _____ of a host cell with master virus for its replication

- A. Structural Unit
- B. protein
- C. Co-infection w/ a helper Virus
- D. cytoplasm

32. The following are the specific examples of diseases caused by prions except -
 A. Kuru
 B. Chronic wasting disease
 C. Conjunctivitis
 D. Scrapie
33. During the replication process, enveloped viruses acquire ----- as the virus buds through the cell membrane.
 A. Nucleic acid
 B. Lipid membrane
 C. Carbohydrates
 D. Viral protein
34. Herpesviridae is the name of
 A. Subfamily
 B. Genera
 C. Family
 D. None of the above
35. Alpha herpesvirinae is the name of a
 A. Subfamily
 B. Genera
 C. Family
 D. None of the above
36. Herpesvirus is the name of
 A. Subfamily
 B. Genera
 C. Family
 D. None of the above
37. Which of the following is not true regarding viruses?
 A. All are obligate intracellular parasites
 B. All have a protein capsid and an envelope
 C. Release virions during cell lysis or budding
 D. All are true
38. Identify the term that describes a disinfectant that can kill virus:
 A. Antiviral
 B. Viricides
 C. Pathogenic
 D. Bacteriosis
39. All the following are nucleoside analogues EXCEPT
 A. ddI (Didanosine)
 B. ddC (Zalcitabine)
 C. Ritonavir
 D. AZT (Zidovudine)
40. The cyclic amine, amantadine, is used in the treatment of:
 A. Influenza virus infections
 B. Herpes virus infections
 C. Parvovirus infections
 D. Poxvirus infections
41. Replication of DNA viruses typically occur in the
 A. Golgi
 B. Cytoplasm
 C. Nucleus
 D. ER
42. How do most viruses acquire envelopes?
 A. By the cell phagocytizing the virus
 B. By generating it themselves
 C. Through glycosylation
 D. By budding through a host cell cytoplasmic membrane